

P28. solution:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow \text{rank}(A) = 1$$

$$\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 1 & 1 & 1 \\ 1 & 1-\lambda & 1 & 1 \\ 1 & 1 & 1-\lambda & 1 \\ 1 & 1 & 1 & 1-\lambda \end{bmatrix}$$

$$= \det \begin{bmatrix} -\lambda & \lambda & 0 & 0 \\ 0 & -\lambda & \lambda & 0 \\ 0 & 0 & -\lambda & \lambda \\ 1 & 1 & 1 & 1-\lambda \end{bmatrix}$$

$$= -\lambda \begin{vmatrix} -\lambda & \lambda & 0 \\ 0 & -\lambda & \lambda \\ 1 & 1 & 1-\lambda \end{vmatrix} - \lambda \begin{vmatrix} 0 & \lambda & 0 \\ 0 & -\lambda & \lambda \\ 1 & 1 & 1-\lambda \end{vmatrix}$$

$$= -\lambda(-\lambda^3 + 3\lambda^2) - \lambda^3$$

$$= \lambda^4 - 4\lambda^3 = \lambda^3(\lambda - 4)$$

$$\Rightarrow \lambda_1 = 4, \lambda_2 = \lambda_3 = \lambda_4 = 0$$

$$C = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \Rightarrow \text{rank}(C) = 2$$

$$C \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \lambda_1 = 2$$

$$C \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \lambda_2 = 2$$

With $\text{rank}(C) = 2$, we know

$$\lambda_3 = \lambda_4 = 0$$

S6.2.

P1. solution:

$$(a) A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 2 \\ 0 & 3-\lambda \end{bmatrix} = (1-\lambda)(3-\lambda) = 0$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = 3$$

$$\lambda_1 = 1, \text{ solve } (A - I)x_1 = 0$$

$$\begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} x_1 = 0 \Rightarrow x_1 = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda_2 = 3, \text{ solve } (A - 3I)x_2 = 0$$

$$\begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} x_2 = 0 \Rightarrow x_2 = c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, S^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$$\det(A - \lambda I) = \det \begin{bmatrix} 1-\lambda & 1 \\ 2 & 2-\lambda \end{bmatrix}$$

$$= (1-\lambda)(2-\lambda) - 2$$

$$= \lambda^2 - 3\lambda = \lambda(\lambda - 3) = 0$$

$$\Rightarrow \lambda_1 = 0, \lambda_2 = 3$$

$$\lambda_1 = 0, \text{ solve } Ax_1 = 0,$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} x_1 = 0 \Rightarrow x_1 = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\lambda_2 = 3, \text{ solve } (A - 3I)x_2 = 0$$

$$\begin{bmatrix} -2 & 1 \\ 2 & -1 \end{bmatrix} x_2 = 0 \Rightarrow x_2 = c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow S = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}, S^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 2/3 & -1/3 \\ 1/3 & 1/3 \end{bmatrix}$$

P2. solution:

$$A^3 = S \Lambda^3 S^{-1}$$

$$A^{-1} = S \Lambda^{-1} S^{-1}$$

P4. solution:

$$A + 2I = S(\Lambda + 2I)S^{-1}$$

$$P5. (a) \text{ False. } (b) \text{ True.}$$

$$(c) \text{ True } (d) \text{ False.}$$

P7. solution:

$$A = \begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix}$$

$$\det(A - \lambda I) = (4 - \lambda)(2 - \lambda) = 0$$

$$\Rightarrow \lambda_1 = 4, \lambda_2 = 2$$

S6.1.

P3. Solution: $A = \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix}$

$$\det(A - \lambda I) = \det \begin{bmatrix} -\lambda & 2 \\ 2 & 3-\lambda \end{bmatrix}$$

$$= -\lambda(3-\lambda) - 4$$

$$= \lambda^2 - 3\lambda - 4$$

$$= (\lambda - 4)(\lambda + 1)$$

$$(\lambda - 4)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 4, \lambda_2 = -1$$

For $\lambda_1 = 4$, solve $(A - 4I)x = 0$

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow x = c \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

For $\lambda_2 = -1$, solve $(A + I)x = 0$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow x = c \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Consider A^{-1}

$$AX = \lambda X \Rightarrow X = \lambda A^{-1}X$$

$$\Rightarrow A^{-1}X = \frac{1}{\lambda}X$$

$\Rightarrow A^{-1}$ has eigenvalues $\frac{1}{\lambda_1} = \frac{1}{4}$

and $\frac{1}{\lambda_2} = -1$. with same set of eigenvectors as that of λ_1 and λ_2 for A .

P6. Solution: $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow \lambda_1 = 1 = \lambda_2$

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow \lambda_1 = 1 = \lambda_2$$

$$AB = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\det(AB - \lambda I) = \det \begin{bmatrix} 1-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}$$

$$= (1-\lambda)(2-\lambda) - 1$$

$$= \lambda^2 - 3\lambda + 1 = 0$$

$$\Rightarrow \lambda = \frac{3 \pm \sqrt{5}}{2}$$

$$BA = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow \det(BA - \lambda I)$$

$$= \lambda^2 - 3\lambda + 1 = 0$$

$$\Rightarrow \lambda = \frac{3 \pm \sqrt{5}}{2}$$

P8. Solution: (a) $\frac{AX}{X}$

(b) solve $(A - \lambda I)x = 0$

P15. Solution:

(a) $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$$\det(P - \lambda I) = \det \begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & 0 & -\lambda \end{bmatrix}$$

$$= -\lambda(\lambda^2) - (-1)$$

$$= -\lambda^3 + 1$$

$$-\lambda^3 + 1 = 0 \Rightarrow \lambda^3 = 1 \Rightarrow \begin{aligned} \lambda_1 &= 1 \\ \lambda_2 &= \frac{-1 + \sqrt{3}i}{2} \\ \lambda_3 &= \frac{-1 - \sqrt{3}i}{2} \end{aligned}$$

(b) $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$$\det(P - \lambda I) = \det \begin{bmatrix} -\lambda & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & -\lambda \end{bmatrix}$$

$$= -\lambda(1-\lambda)(-\lambda) - (1-\lambda)$$

$$= \lambda^2(1-\lambda) - (1-\lambda)$$

$$= (1-\lambda)(\lambda^2 - 1)$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = 1, \lambda_3 = -1$$

P19. Solution: (a), (b), (d)

$$\lambda_1 = 4, \text{ solve } (A - 4I)x_1 = 0$$

$$\begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix} x_1 = 0 \Rightarrow x_1 = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 2, \text{ solve } (A - 2I)x_2 = 0$$

$$\begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix} x_2 = 0 \Rightarrow x_2 = c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow S = \begin{bmatrix} 2c_1 & 0 \\ c_1 & c_2 \end{bmatrix} \text{ for all } \begin{matrix} c_1 \neq 0 \\ c_2 \neq 0 \end{matrix}$$

$$\text{with } A = S \begin{bmatrix} 4 & \\ & 2 \end{bmatrix} S^{-1}$$

P11. Solution:

$$S = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}, S^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$$

$$S \Lambda^k S^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = S \Lambda^k \begin{bmatrix} 1/(\lambda_1 - \lambda_2) \\ -1/(\lambda_1 - \lambda_2) \end{bmatrix}$$

$$= S \begin{bmatrix} \frac{\lambda_1^k}{\lambda_1 - \lambda_2} \\ \frac{\lambda_2^k}{\lambda_1 - \lambda_2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\lambda_1^{k+1} - \lambda_2^{k+1}}{\lambda_1 - \lambda_2} \\ \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2} \end{bmatrix}$$

$$\Rightarrow F_k = \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2}$$

P19. Solution: 1

$$A = \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix}$$

$$\det(A - \lambda I) = \det \begin{bmatrix} 0.6 - \lambda & 0.4 \\ 0.4 & 0.6 - \lambda \end{bmatrix}$$

$$= \lambda^2 - 1.2\lambda + 0.2$$

$$= (\lambda - 1)(\lambda - 0.2)$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = 0.2$$

$$\Rightarrow A^k \rightarrow 0$$

$$B = \begin{bmatrix} 0.6 & 0.9 \\ 0.1 & 0.6 \end{bmatrix}$$

$$\det(B - \lambda I) = \det \begin{bmatrix} 0.6 - \lambda & 0.9 \\ 0.1 & 0.6 - \lambda \end{bmatrix}$$

$$= \lambda^2 - 1.2\lambda + 0.27$$

$$= (\lambda - 0.9)(\lambda - 0.3)$$

$$\Rightarrow \lambda_1 = 0.9, \lambda_2 = 0.3 \Rightarrow A^k \rightarrow 0$$

P23. Solution:

$$B = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\det(B - \lambda I) = (3 - \lambda)(2 - \lambda) = 0$$

$$\Rightarrow \lambda_1 = 3, \lambda_2 = 2$$

$$\lambda_1 = 3, \text{ solve } (A - 3I)x_1 = 0 \Rightarrow \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} x_1 = 0$$

$$x_1 = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda_2 = 2, \text{ solve } (A - 2I)x_2 = 0 \Rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} x_2 = 0$$

$$\Rightarrow x_2 = c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow S = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}, S^{-1} = - \begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$\Rightarrow B = S \begin{bmatrix} 3 & \\ & 2 \end{bmatrix} S^{-1}$$

$$B^k = S \begin{bmatrix} 3 & \\ & 2 \end{bmatrix}^k S^{-1}$$

$$= \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3^k & \\ & 2^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3^k & 3^k - 2^k \\ 0 & 2^k \end{bmatrix}$$